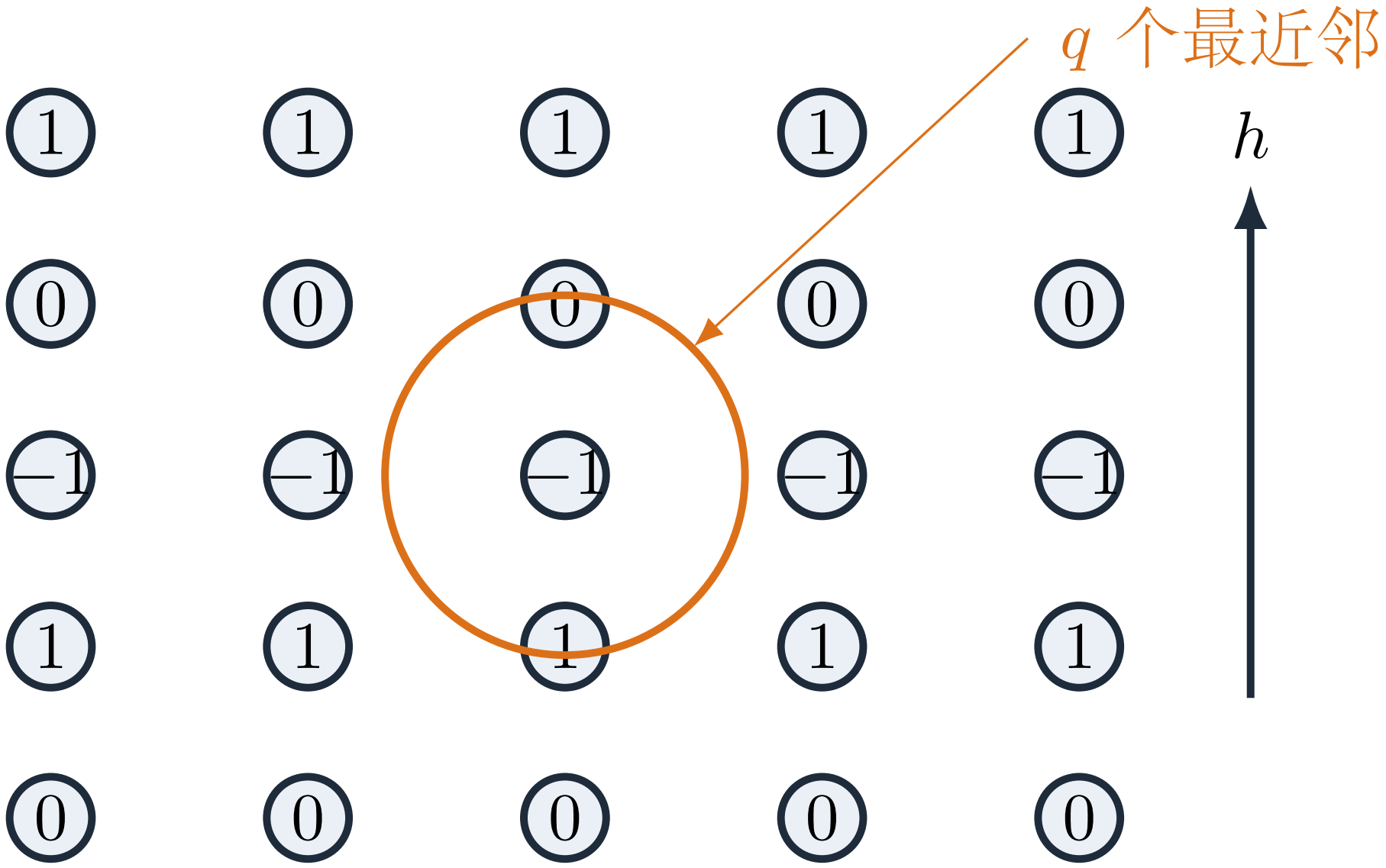

$$S_z \in \{-1, 0, +1\}$$